Task 1: Geocoding Office Data

Whole US Implementation

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Abstract

This report summarizes the geocoding pipeline for CoStar commercial real estate lease data. The pipeline standardizes addresses and submits them to the U.S. Census Bureau Batch Geocoder API to obtain latitude, longitude, census tract, and FIPS codes. We first validated the approach on San Francisco (9,352 properties, 96% match rate), then scaled it to the entire U.S. dataset of 1,051,219 properties, achieving a 92.96% geocoding success rate. We additionally evaluated ZIP code matching as a post-geocoding validation step and found it does not materially improve outcomes.

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1 This Week's Progress

The geocoding pipeline is now complete for the entire U.S. dataset. All 1,051,219 unique commercial properties from the CoStar lease database have been processed through address standardization, Census API geocoding, and ZIP code validation. The pipeline ran across 11 batches to stay within file size constraints and Census API limits.

Earlier in the project, we piloted the approach on San Francisco, achieving a 96% geocoding match rate across 9,352 properties. We then met to align on standardization methodology and confirmed our approach was consistent with reference code. The SF pilot confirmed the pipeline was ready for national scale.

2 Address Standardization

The first step was to confirm we are using the same standardization process as reference code (Max's Code). After reviewing the reference, our standardization was determined to be an improvement.

Our standardization procedure converts addresses to uppercase, removes punctuation (except commas), abbreviates street types (e.g., STREET to ST, AVENUE to AVE), and collapses address ranges (e.g., "1234-5678 Main St" becomes "1234 Main St"). This format matches what the Census Batch Geocoder expects.

3 ZIP Code Matching as a Validation Strategy

After geocoding, we evaluated ZIP code matching as a post-geocoding validation step. The idea was to compare the input ZIP code (from CoStar) against the ZIP code returned by the Census geocoder, and use agreement as a signal of match quality.

3.1 San Francisco Results

Table 1 shows the ZIP match breakdown for the SF pilot (9,352 properties).

Table 1: San Francisco ZIP Match Detail Counts

zip_match_detail	rows	pct
One-to-One Matched	8,937	95.56
One-to-One Did Not Match	42	0.45
One-to-Many Matched Exactly One Output	0	0.00
One-to-Many Matched Multiple Outputs	12	0.13
One-to-Many Matched No Output	6	0.06
Geocoder ZIP Missing	354	3.79
Original ZIP Missing	1	0.01

3.2 Whole U.S. Results

Table 2 shows the same breakdown scaled to the full U.S. dataset (1,051,219 properties).

Table 2: Whole U.S. ZIP Match Detail Counts

zip_match_detail	rows	pct
One-to-One Matched	886,635	84.34
One-to-One Did Not Match	6,977	0.66
One-to-Many Matched Exactly One Output	15	0.00
One-to-Many Matched Multiple Outputs	5,969	0.57
One-to-Many Matched No Output	306	0.03
Geocoder ZIP Missing	67,721	6.44
Original ZIP Missing	83,596	7.95

3.3 Assessment

ZIP-based matching does not meaningfully improve geocoding outcomes. Even among records that appear to match at first glance, the approach introduces additional complexity, extra columns requiring flagging and review. In one-to-many cases with multiple outputs, we still observe a single input ZIP mapping to multiple candidate ZIPs or to incorrect ZIPs. The high rate of missing ZIP codes (6.44% from the geocoder, 7.95% from the original data) further limits the utility of this approach. A census-tract matching strategy is also unlikely to materially improve outcomes for the same reasons.

Fix

4 Overall Match Statistics

Table 3 shows the geocoding match statistics across the entire U.S. dataset.

Table 3: Whole U.S. Geocoding Overall Summary

Census track cond No match?

Metric	Value
Total Properties	1,051,219
Matched	970,482
Tied (Multiple Candidates)	6,697
Matched or Tied	977,179
Strict Match Rate	92.32%
Matched-or-Tied Rate	92.96%

text No Match

Table 4 shows the breakdown by match type.

Table 4: Whole U.S. Match Type Counts

Fix

match_type	rows	pct
One-to-One	1,044,522	99.36
One-to-Many (Tie)	6,697	0.64

Figure 1 shows the strict match rate geographically. Table 5 shows the full breakdown by state, sorted by strict match rate descending.

Sub file

each CO stats

tied multiple returns quick stat check

Check CTA

Add look at Image 1

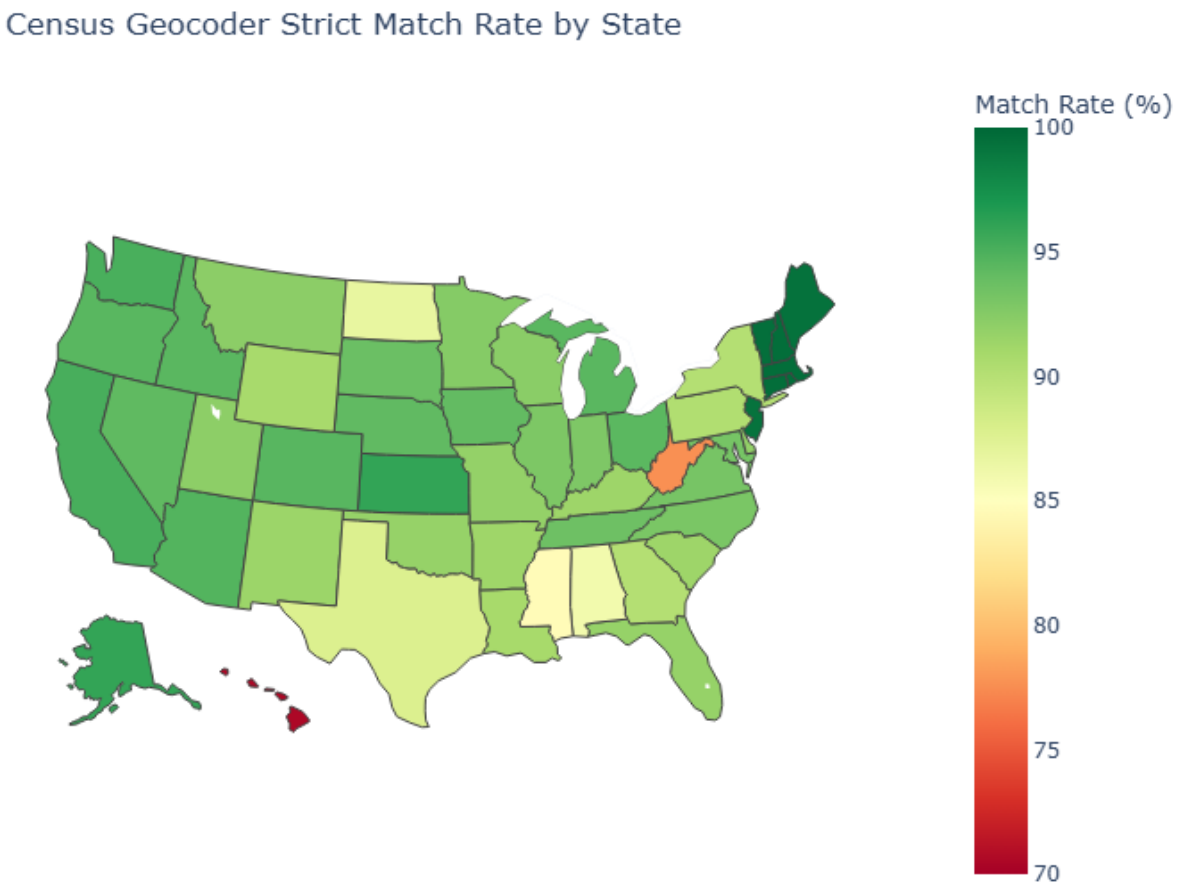

Figure 1: Census Geocoder Strict Match Rate by State

Table 5: Match Rate by State

State	Total	Matched	Tied	Strict Match Rate (%)	Match-or-Tie Rate (%)
Rhode Island	3,667	3,654	13	99.65	100.00
New Hampshire	3,782	3,768	14	99.63	100.00
Massachusetts	21,281	21,199	82	99.61	100.00
Connecticut	14,131	14,075	56	99.60	100.00
Vermont	1,101	1,096	5	99.55	100.00
Maine	2,975	2,957	18	99.39	100.00
New Jersey	30,168	29,957	211	99.30	100.00
District of Columbia	3,291	3,187	3	96.84	96.93
Kansas	7,596	7,296	32	96.05	96.47
Alaska	1,058	1,016	3	96.03	96.31
California	165,793	157,804	656	95.18	95.58
Washington	23,906	22,746	132	95.15	95.70
Arizona	24,670	23,373	60	94.74	94.99
Colorado	22,600	21,368	81	94.55	94.91
Oregon	14,913	14,093	81	94.50	95.04
Michigan	31,221	29,496	151	94.47	94.96

State	Total	Matched	Tied	Strict Match Rate (%)	Match-or-Tie Rate (%)
Idaho	5,704	5,388	32	94.46	95.02
Ohio	37,188	35,120	213	94.44	95.01
Nevada	11,878	11,194	50	94.24	94.66
Nebraska	4,953	4,666	17	94.21	94.55
Iowa	5,690	5,356	49	94.13	94.99
South Dakota	1,602	1,502	10	93.76	94.38
Tennessee	19,188	17,968	184	93.64	94.60
Maryland	18,351	17,116	77	93.27	93.69
Virginia	25,455	23,723	79	93.20	93.51
North Carolina	34,365	31,982	275	93.07	93.87
Illinois	39,768	36,959	310	92.94	93.72
Indiana	17,857	16,587	156	92.89	93.76
Minnesota	13,224	12,239	113	92.55	93.41
Wisconsin	13,646	12,598	70	92.32	92.83
Montana	1,959	1,807	24	92.24	93.47
Utah	10,044	9,256	56	92.15	92.71
Delaware	2,954	2,714	18	91.88	92.48
Missouri	17,780	16,327	179	91.83	92.83
Oklahoma	12,095	11,099	66	91.77	92.31
Florida	79,864	73,271	809	91.74	92.76
New Mexico	5,291	4,840	48	91.48	92.38
South Carolina	19,076	17,431	180	91.38	92.32
Kentucky	9,212	8,418	78	91.38	92.23
Arkansas	6,428	5,872	45	91.35	92.05
Louisiana	10,206	9,275	79	90.88	91.65
Wyoming	722	656	3	90.86	91.27
Pennsylvania	37,662	34,021	213	90.33	90.90
Georgia	36,736	33,120	479	90.16	91.46
New York	62,311	56,172	354	90.15	90.72
Texas	88,171	77,391	556	87.77	88.40
North Dakota	1,748	1,517	31	86.78	88.56
Alabama	11,378	9,789	141	86.03	87.27
Mississippi	4,320	3,655	77	84.61	86.39
West Virginia	2,724	2,114	17	77.61	78.23
Hawaii	3,197	2,254	11	70.50	70.85

5 Appendix

Link to code repository:

- https://github.com/WilliamClintC/Geocoding_Office